

An Online Therapeutic Intervention for Veterans Patients Suffering with Chronic Pain

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ABSTRACT

Chronic pain affects a large proportion of veterans. Initially, the solution to this medical illness was providing pharmacological medical prescriptions. However, opioid-based medication can lead to overuse or misuse. Thus, the Veteran's Health Administration (VHA) is interested in non-pharmacological approaches that target both pain management and chronic pain-related functional issues. A solution adopted by the VHA is providing Acceptance and Commitment Therapy (ACT) for chronic pain veterans. This manuscript describes a project to develop and evaluate Veteran ACT for Chronic Pain (VACT-CP), an online therapy conducted by an embodied conversational agent (ECA) portraying a mental healthcare coach tasked to lessen the effects of chronic pain and improve functioning.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Chronic pain affects a large percentage of the 20 million United States Veterans according to the Department of Veterans Affairs (VA) [29]. Chronic Pain (CP) is defined as pain lasting for more than 12 weeks beyond the initial injury and is often associated with depression, anxiety, sleep deprivation, decreased quality of life, and substance use disorder. It is also a risk factor for suicide. The consequences of chronic pain include a decrease in work productivity and an increase in healthcare costs. Chronic pain is particularly problematic in U.S. Veteran populations, as a study revealed that 44% of 2,500 surveyed veterans suffered from chronic pain, and also this illness is more prevalent and of greater intensity in veterans than in the general population [5, 36].

The VA provides a wide variety of treatments to alleviate veterans that suffer from chronic pain. Medication is often prescribed for these patients, but unfortunately use and misuse of many medications can cause addiction or side effects that lead to overdose, hospitalization, and death. Moreover, data shows that 70% of homeless veterans also experience a substance use disorder [4]. Therefore, the VA considers non-opioids alternative therapeutic approaches for treating pain management such as Cognitive Behavioral Therapy, Mindfulness-based Stress Reduction, and Acceptance and Commitment Therapy [3].

Acceptance and Commitment Therapy (ACT) is a promising treatment for patients with chronic pain. ACT is an empirically-based practice offered across the US national VA system for the treatment of depression. The goal of ACT therapy is to accept our difficult or uncomfortable thoughts, feelings, and physical sensations in pursuit of a rich and meaningful life. This involves helping clients to build psychological flexibility, which includes six components:

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acceptance, defusion, self-as-context, values, committed action, and contact with the present moment. ACT is both an empirically-based and flexible psychotherapy that allows the therapist to create and individualize their own mindfulness techniques, or even to co-create them with the patients [19]. Despite the evidence suggesting that ACT is an effective intervention to treat chronic pain [13, 25, 40, 42], not all mental health specialists in the VA are trained or available to implement it [15].

Embodied Conversational Agents (ECA) could be a valid instrument for delivering ACT therapy to patients through online digital platforms. ECAs are synthetic entities that mimic human appearance and behavior. They simulate properties of human face-to-face conversation by recognizing, responding, and generating verbal and nonverbal behaviors, and engaging in conversational functions (turn-taking, feedback, and repair mechanisms) [10]. Furthermore, ECAs can be highly personalized and tailored to fit any particular demographic group such as gender, race, or age group. Additionally, ECAs could be deployed in different formats such as standard websites, virtual reality systems, serious games, or stand-alone software. The medical field employs ECAs for training medical staff [39], teaching medical-related interviewing skills [32], or promoting physical activity [41].

The clinical psychology field has begun to utilize ECAs for psychotherapeutic interventions. Mental-healthcare interventions with ECAs include, but are not limited to, Social skills training, Educational Aid, Cognitive Behavioral Therapy, Counseling, or Tutoring [30]. A study portrayed an Embodied Conversational Agent as a virtual counselor to elicit active soldiers and war veterans to self-disclose their personal problems. Overall, evidence shows that ECAs can have a positive effect on user engagement and involvement and that they can be effective in this sense as an adjunct to already existing interventions for mental disorders.

In this manuscript, we describe a therapeutic intervention for veterans that suffer from chronic pain. Patients from the U.S. Veterans Affairs participated in web-based Veteran ACT for chronic pain (VACT-CP) sessions with an embodied Conversational Agent portraying the role of a healthcare Coach. In this seven-week module program, patients engaged in Acceptance and Commitment (ACT) therapy sessions with a mental healthcare ECA, named Coach Anne. During this intervention, we examined qualitative changes in the patient's functioning, quality of life, and system usability.

2 RELATED WORK

2.1 Chronic Pain

Chronic pain (CP) is pain that lasts between 3 to 6 months, or longer [1]. The National Institutes of Health (NIH), reported that 65.6% of American Veterans suffered from severe pain in the three months before they were surveyed [28]. Additionally, the VA revealed that chronic pain prevalence is high among Veterans and that 1 in 3 Veterans have been diagnosed with a condition related to chronic pain [2]. Primary care facilities provide several treatments for patients that suffer from CP. These treatments include pharmacological treatments, surgery, or implantable drug delivery systems [37]. The VA is exploring different non-pharmacological strategies since in

some cases these may not be effective in reducing chronic pain [27] and may lead to worse outcomes by decreasing pain thresholds [12] or contributing to other adverse effects [8]. On the contrary, ACT is a promising coping strategy associated to control and diminish the role of pain in the patient's daily activity [31].

2.2 Acceptance and Commitment Therapy for Chronic Pain

The core principles of ACT therapy rely on a framework named Psychological Flexibility (PF) model. An overall description of the PF framework is the capacity to be directly, consciously, and fully in contact with the present moment and urges the individual to change their relationships with private events (i.e., thoughts, memories, feelings, and bodily sensations) and not the events themselves [43]. The PF model includes six interrelated core processes namely *Acceptance, Cognitive defusion, being present, self as context, values, and committed action* [21]. The ultimate goal of this framework is to initiate and maintain the health behavior change. ACT therapy showed successful outcomes with patients that suffer from chronic pain [22–24] and robust evidence for increasing functioning and decreasing opioid dosage [14]. Internet-based ACT for chronic pain can increase activity engagement and pain acceptance while decreasing pain-related distress and comorbid mood disorders [9, 38].

2.3 ECA in Therapeutic applications

Embodied Conversational Agents (ECAs) have an increasingly important role in healthcare. Especially, in scenarios that demand highly social and conversational interaction such as mental therapy [6]. Additionally, software and hardware advances and accessibility allow for more complex dialogue management methods and more conversational flexibility [26]. Conversational agents can leverage their capabilities to establish rapport, trust, and therapeutic alliance with patients. These attributes can be applied in the context of therapeutic relationships between mental healthcare counselors and patients. ECAs can assist clinicians during the consultation, support consumers with behavior change challenges, or assist patients and elderly individuals in their living environments. Positive outcomes in mental health therapies show the effectiveness of ECAs. Thus, ECAs are successfully implemented for treating anxiety disorders and depression [16], for unguided cognitive behavioral therapy [35], treatment of phobias [17], or mental well-being [18].

3 SYSTEM DESCRIPTION

3.1 Simulation Setup

This manuscript outlines our intention to develop and evaluate a platform named Veteran ACT for Chronic Pain (VACT-CP). This platform is an online program guided by an embodied conversational agent coach (ECA) designed to improve pain management and functioning in veterans that suffer from chronic pain.

The therapeutic intervention was developed in the Unity game engine and rendered as a WebGL application deployed on a private server. This simulation can be accessed by any standard web browser such as Firefox or Google Chrome. The simulation is highly interactive allowing users to write their own custom goals, report their subjective impressions, and select their preferences using menus and check-boxes. These variables created during the sessions are recorded and are included in the later stages of the therapy by Coach Anne. The ECA utterances, the patient's responses, custom goals, and overall objective metrics such as time in the simulation are stored in an SQL database. This information is used to tailor the therapy for each patient, keep track of the progress of the patient in the therapy, and for statistical analysis at the end of the intervention. Finally, the system uses a wide variety of formats to present information to the patient such as video material of patient testimonials, audio, and images.



Figure 1: Coach Anne conducting a therapy session. The image shows the graphical user interface menu displaying the available options on the Main Menu.

3.2 Mental Healthcare Coach Anne

The system uses a hierarchical task-network-based dialogue manager to drive the interaction between a user and the agent. After each utterance is produced by the agent, a graphical user interface is displayed on the screen allowing users to select an appropriate response from multiple choices (See Figure 1). Participants interacted with a conversational agent that portrayed the mental healthcare coach named Anne (see Figure 1). In each therapy session, the ECA leverages verbal and nonverbal behaviors to persuade users toward health behavior change. These behaviors include hand gestures, gaze cues, eyebrow raises, and body posture shifts, generated by the BEAT text-to-embodied speech engine [11]. These behaviors are carefully displayed according to each conversation. For example, in case the user was able to complete a weekly goal, coach Anne congratulates the patient verbally, smiles, and displayed a thumbs-up gesture. On the contrary, if the user was not able to achieve a goal, the ECA remains serious or expresses concern by frowning or showing a neutral facial expression. These behaviors are critical since the goal of this intervention is to elicit a behavioral change and attitude.

3.3 Veteran ACT for Chronic Pain (VACT-CP) System

The goal of this online therapeutic intervention, named Veteran ACT for Chronic Pain (VACT-CP), is to guide veterans with chronic pain to decrease behavioral avoidance and increase pain acceptance and valued living to improve their functioning. The system was designed following advice from mental healthcare specialists and clinical professionals who treat chronic pain. Patients engaged in therapy with the ECA following a modular design. Patients accessed a total of seven modules on a week-by-week basis, with one additional module on mindfulness available at any time proposed by Hayes [20]. Table 1 shows a description of each module.

4 STUDY DESIGN

A pilot testing was conducted virtually using a secure videoconferencing platform or in person at the Bedford VAMC facility if preferred by the participant. The pilot test included three waves with 4-5 participants per wave. The reasoning behind having 3 different usability testing waves, with at least one week in between, was to allow for time to iteratively address any functionality issues or concerns with the user interface raised by the participants in the interviews.

4.1 Measures

For this pilot evaluation, we included the Brief Pain Inventory (BPI) [33], the technology acceptance model (TAM) [34], and the System

Table 1: Shows each of the modules of the VACT-CP Therapy.

TIMELINE	FOCUS	INTERVENTION
Week 1	Introduction	The introductory module assesses the nature of the veteran's pain, past treatments, and experience with acceptance and mindfulness. Coach Anne provides information on ACT and assesses the veteran's values and goals to help them set a weekly goal.
Week 2	Behavioral Change and Triggers	This module focuses more broadly on pain psychoeducation and pain management. Veterans complete exercises on dealing with cognitive barriers with the goal of changing their own self-identified obstructions to living a meaningful life.
Week 3	Acceptance & Mindfulness	This module uses metaphors and a veteran narrative story to explain and explore concepts of acceptance and mindfulness, and the veteran completes an acceptance exercise linked to previously-identified values.
Week 4	Cognitive Defusion	This module focuses on the ACT concept of cognitive defusion using psychoeducation about the nature of language and cognitive barriers to valued living. Metaphors, mindfulness, and ACT exercises are included.
Week 5	Willingness	This module explains the ACT concept of willingness and discusses willingness to experience discomfort as a means to pursuing important valued goals. The veteran creates and applies a willingness hierarchy to identify a goal for the week that embodies values-driven willingness.
Week 6	Committed Action	This module focuses on the ACT concept of committed action. Coach Anne helps the user identify personal barriers to committed action and strategies for dealing with these perceived barriers.
Week 7	Valued Living Wrap-up	The final session provides an overview of ACT processes, a review of goals achieved, and reinforces progress made during the program. This session ends with planning for the "lifelong assignment" of fully engaging in one's life.
Additional Modules	Mindfulness Exercises	This module introduces the topic of mindfulness and provides a series of questions to route users to use exercises that align with their goal for the mindfulness experience. This module is accessible anytime after completing the first module.
	Additional Resources	This module provides additional information, links, and resources for care options within the Veteran's Health Administration for pain, psychiatric conditions, and physical health management.

Usability Scale (SUS) [7]. For the pain measures we present the Pain severity and Pain Interference (scale: 0-10); for the TAM we show the General Positive Attitude Toward Tech (scale: 0-10) and the General Tech Dependence/Anxiety (scale: 1-10). Additionally, for the Usability Measures we used the System Usability Scale (scale 0-100), Ease of Use, Motivation for Future Use, and Perceived Usefulness (scale: 0 to 4).

4.2 Procedure

Each patient first completed a phone screen to assess whether veterans are eligible according to our inclusion/exclusion criteria. After the intake assessment, eligible participants scheduled a single 2-hour usability testing session. After providing verbal informed consent, participants completed a pre-usability testing survey battery. Next, they accessed the VACT-CP website with credentials created for them and they completed the first and second modules of the VACT-CP system. Patients were instructed to "think aloud" as they complete the modules. At the end of the seventh module, patients accessed an online form to complete surveys to assess their subjective impressions of the therapy. Participants received a \$40 gift card for their 2-hour virtual research session.

4.3 Results

Descriptive analysis in Table 2 shows the results from three different constructs namely, Pain Measures, Technology Acceptance Measures (TAM), and Usability Measures. Finally, the population sample was too small to conduct a statistical analysis. However, the mean scores show a decrease in the pain-related constructs from Wave 1 to Wave 3 while the usability mean scores increase.

Table 2: shows mean values by each testing Wave.

	Wave 1	Wave 2	Wave 3
Pain Severity	6.94	5.81	6.85
Pain Interface	6.36	6.32	6.61
Gral. Positive Attitude Toward Tech	7.83	6.79	8.23
Gral. Tech Dependence/Anxiety	5.75	2.25	6.73
System Usability Scale	73.13	80	80.5
Ease of Use	2.75	3.25	3.53
Motivation for Future Use	2.58	2.59	2.94
Perceived Usefulness	2.38	2.81	3.15

5 DISCUSSION

The Covid-19 pandemic exacerbated the need to develop online programs for patients to avoid visits to the doctor and hospital during the lockdown. The intended outcome product, the online Veteran ACT for Chronic Pain (VACT-CP), requires this multi-phase project to successfully apply user-centered design best practices in developing our proposed intervention.

6 CONCLUSION AND LIMITATION

This manuscript describes a web-based therapeutic intervention named online Veteran ACT for Chronic Pain. The therapy was conducted by an ECA during seven modules in seven weeks. The ECA provides coping techniques to better manage the impact that mental or physical chronic pain difficulties have on their daily activities. Currently, this study is actively recruiting participants from the U.S. Veterans Affairs department. There are limitations of this study specifically related to its small sample size and therefore, generalizability. In addition, the format of the website requires a person to have access to a laptop or desktop computer, as the current website is not mobile-enabled.

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